

Def: ($G\text{Top}$)

$G\text{Top}$ is a category.

• Objects: G -spaces. G -space: a space X with a conti map $\alpha: G \times X \rightarrow X$, G with discrete top.

• Morphism: G -equivariant maps: $f: X \rightarrow Y$ with $f(g \cdot x) = g \cdot f(x)$.

Exp:

Antipodal action of C_2 on S^n . This is a free action. $\rightarrow$ no fixed point

Let V be a rep of G . Denote by S^V the one-point compactification of V . If V is an n -dim rep, then S^V is an n -dim sphere.

$$\text{For } g \in G, x \in S^V, g \cdot x = \begin{cases} g \cdot x, & x \in V \\ \infty, & x = \infty \end{cases}$$

$S^{(-)}$ does not define a functor $\text{Mod}_{\mathbb{R}[G]} \rightarrow G\text{Top}$. A map $V \rightarrow W$ of reps is proper iff it is injective. So we get a functor

$$S^{(-)}: \text{Mod}_{\mathbb{R}[G]}^{\text{inj}} \rightarrow G\text{Top}$$

$\hookrightarrow$ the subcategory of injective maps of reps.

Def: (based G -space)

A based G -space is a G -space X equipped with an equi map $*$ $\rightarrow$ X . A map of based G -spaces is just a basepoint-preserving that is equi.

Denote by $G\text{Top}_*$ the category of based G -spaces.

Def (induced G -space)

$H \leq G$. X is an H -space. Define the induced G -space of X up to G , denoted $\uparrow_H^G(X)$ or $G \times_H X$, as the quotient of $G \times X$ by $(g \cdot h, x) \sim (g, h \cdot x)$.

Exp

$$H \leq G. \uparrow_H^G(\ast) = G \times_H \ast \cong G/H.$$

If X is any space with trivial H -action, $\uparrow_H^G(X) = G/H \times X$, where G acts on G/H .

Prop

$H \leq G$. Induction of spaces is left adjoint to restriction.

$$\uparrow_H^G : H\text{-Top} \rightleftarrows G\text{-Top} : \downarrow_H^G$$

$$\text{Map}_G(G \times_H X, Y) \cong \text{Map}_H(X, \downarrow_H^G Y)$$

For G -equiv map $F: G \times_H X \rightarrow Y$, it is decided by values on $[e, x]$, since

$[g, x] = g[e, x]$. Define $f: X \rightarrow \downarrow_H^G Y$, $f(x) = F([e, x])$. f is H -equiv.

For H -equiv map $f: X \rightarrow \downarrow_H^G Y$, define $F: G \times_H X \rightarrow Y$ by $F([g, x]) = g \cdot f(x)$.

Def (induced based G -space)

$H \leq G$. X be a based H -space. Define the induced based G -space of X up to G , denote $\uparrow_H^G(X)$, as $G_+ \wedge_H X$, which denote the quotient of $\underbrace{G_+ \wedge X}_{\downarrow}$ by $(g \cdot h, x) \sim (g, h \cdot x)$.
 $(G \times X) / (G \times x_0)$

Prop

$H \leq G$. Induction of spaces is left adjoint to restriction.

$$\uparrow_H^G: H\text{Top}_* \rightleftarrows G\text{Top}_* : \downarrow_H^G$$

$$\text{Map}_G(G_+ \wedge_H X, Y) \cong \text{Map}_H(X, \downarrow_H^G Y)$$

Def (fixed points)

For $X \in G\text{Top}$, define the fixed points of X to be

$$X^G = \{x \in X \mid g \cdot x = x \text{ for } \forall g \in G\}$$

equipped with the subspace top as a subspace of X .

Ex

$$(S^0)^{C_2} \cong S^0, (S^{C_2})^{C_2} \cong S^1.$$

Prop

The fixed points functor is right adjoint to the trivial G -action functor.

$$\text{triv} = \eta^*: \text{Top} \rightleftarrows G\text{Top} : (-)^G$$

$$\text{Map}_{G\text{Top}}(\text{triv } X, Y) \cong \text{Map}_{\text{Top}}(X, Y^G)$$

Def (H-fixed points functor)

For a G -space, define H -fixed points functor as the composite

$$G\text{Top} \xrightarrow{\downarrow_H^G} H\text{Top} \xrightarrow{(-)^H} \text{Top}$$

Ex

$$G = K_4, (S^{P_1^* \sigma})^L \cong (S^{P_1^* \sigma})^D \cong S^0, (S^{P_1^* \sigma})^R \cong S^1.$$

Def (homotopy)

Let $X \xrightarrow{f} Y$ be equi maps between G -spaces. Letting G act on the interval trivially, a homotopy between f and g is an equi map $h: I \times X \rightarrow Y$ that restricts to f and g at 0 and 1.

This forces each map $h_t: X \rightarrow Y$ to be equi.

Def (equi homotopy group system)

$$\underline{\pi}_n(X, x_0)(G/H) = \pi_n(X^H, x_0), \quad x_0 \in X^G.$$

$$\underline{\pi}_n(X)(G/H) = \pi_n(X^H)$$

$\mathcal{O}_G$ is the orbit category of G :

- objects: $G/H, H \leq G$.
- morphisms: G -equi maps $G/H \rightarrow G/K$

So $\underline{\pi}_n(X): \mathcal{O}_G^{\text{op}} \rightarrow \text{Groups}$ for $n \geq 1$. If $n=0$, images are on sets.

A G -map $\alpha: G/H \rightarrow G/K$ decided by $\alpha(eH) = aK$, s.t. $\alpha^{-1}Ha \leq K$. Then it induces a map $\alpha^*: X^K \rightarrow X^H, x \mapsto a \cdot x$.

For $\underline{A}, \underline{B}: \mathcal{O}_G^{\text{op}} \rightarrow \text{Groups}$ to be two system, a morphism $\eta: A \rightarrow B$ is a natural transformation, s.t.

$$\begin{array}{ccc} \underline{A}(G/K) & \xrightarrow{\eta_K} & \underline{B}(G/K) \\ \downarrow \alpha^* & \curvearrowright & \downarrow \alpha^* \\ \underline{A}(G/H) & \xrightarrow{\eta_H} & \underline{B}(G/H) \end{array}$$

Def (weak G -homotopy equivalence)

A G -equiv map $f: X \rightarrow Y$ is a weak G -homotopy equivalence if it induces an iso on π_n for all $n \geq 0$. It induces a bijection on π_0 and iso from $\pi_n(X, x_0) \rightarrow \pi_n(Y, f(x_0))$, $n \geq 1$, for each choice of basepoint $x_0 \in X^G$.

We have

$$[(G/H)_+ \wedge S^n, X]_G \cong [S^n, X^H] \cong \pi_n(X^H)$$

$\hookrightarrow G$ -equiv homotopy class

For a G -map $F: (G/H)_+ \wedge S^n \rightarrow X$, define $f: S^n \rightarrow X$ by $f(s) = F(H, s)$. We can check $f: S^n \rightarrow X^H$. For a map $f: S^n \rightarrow X^H$, define $F: (G/H)_+ \wedge S^n \rightarrow X$ by $F(gH, s) = g \cdot f(s)$. We can check F is well-defined and F is a G -map.

So

$$\text{Map}_G((G/H)_+ \wedge S^n, X) \cong \text{Map}(S^n, X^H)$$

Then, we get $[(G/H)_+ \wedge S^n, X]_G \cong [S^n, X^H]$.

That means that $(G/H)_+ \wedge S^n$ can detect $\pi_n(X^H)$.

Def: (G -CW complex)

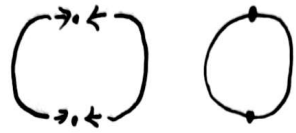
A G -CW complex is a space X built as a colimit of G -space X_n , where X_n is a discrete G -set and X_n is obtained from X_{n-1} by attaching cells of type $(G/H) \times D^n$ (H is not fixed).

Prop:

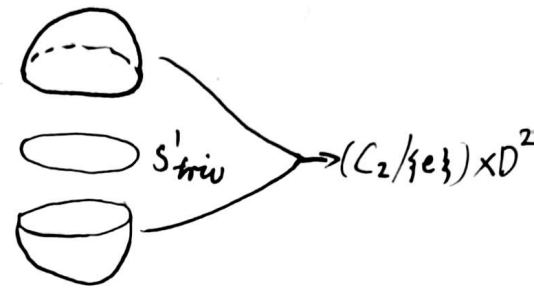
If X is a non-equiv CW-complex, we may consider it as a

G -CW complex with trivial G -action. All cells will be of form $(G/G) \times D^n$.

$G = C_2$. 0-cell s^0 $\vdots$ 1 cell $(C_2/\{e\}) \times D^1$



$G = C_2$. Consider the regular rep sphere, $S^{P_{C_2}}$.
 $X_1 = S'_{triv}$.



Thm (equiv Whitehead Theorem)

Let $f: X \rightarrow Y$ be a weak G -homotopy equivalence between G -CW complexes. Then, f is a G -homotopy equivalence.

Def (universal free G -space, classifying space of G)

G is a top group. A space EG is called a universal free G -space if

- EG is contractible,
- G acts freely on EG .

The classifying space of G , denoted by BG , is defined to be the orbit space

$$BG := EG/G.$$

Equivalently, there is a principal G -bundle

$$G \rightarrow EG \rightarrow BG$$

called the universal principal G -bundle.

exp

$G = C_2$. $EG = S^\infty$ with antipodal action of C_2 . $BC_2 \cong S^\infty/C_2 \cong \mathbb{R}P^\infty$.

$EG \times EH = E(G \times H)$. $B(G \times H) \cong BG \times BH$, for any $G, H \in \text{Groups}$.

loop

The space BG is a $K(G, 1)$ meaning that $\pi_1(BG) \cong G$ and all other homotopy groups vanish.

loop

The cellular chains $C_*(EG)$ on EG form a free resolution of $\mathbb{Z}$ over $\mathbb{Z}[G]$.

Cor

For any abelian group A , thought of as a trivial G -module, we have isos $H_*(BG; A) \cong H(G; A)$ and $H^*(BG; A) \cong H^*(G; A)$ $\hookrightarrow \mathbb{Z}[G]$ -module

Def: (Borel construction)

Given a G -space X , define the Borel construction on X to be the orbit space of the G -action on $EG \times X$. The Borel construction is often denoted $EG \times_G X$ and we define the Borel-equivariant homology and cohomology of X to be

$$H_*^{\text{Borel}}(X) = H_*(EG \times_G X), H_*^{\text{Borel}}(X) = H^*(EG \times_G X)$$

$$EG \times_G X := (EG \times X)/G.$$

The G -action on $EG \times X$ is $g \cdot (e, x) = (ge, gx)$, $g \in G, (e, x) \in EG$.

$X_{hG} := EG \times_G X \rightarrow$ It is also called the homotopy orbit space.

Exap $X^{hG} := \text{Map}_G(\bar{E}G, X)$, the homotopy fixed point space

a C_2 -equiv map: $S_a' \rightarrow S'$
 $\downarrow$ trivial action
 antipodal action



If $\cdot 2$ is an iso in a abelian group A ,

$$H_{\text{Borel}}^*(S'; A) \xrightarrow{\sim} H_{\text{Borel}}^*(S_a'; A)$$

is an iso. But S' and S_a' are not C_2 -equiv homotopy equivalent, since their fixed points spaces are not equivalent.

So Borel homology may not tell the difference between fixed point spaces.

Def (underlying equivalence)

A G -equiv map $f: X \rightarrow Y$ is an underlying equivalence if it induces an equivalence on underlying spaces.

Prop The functor $\bar{E}G \times (-): G\text{Top} \rightarrow G\text{Top}$ takes underlying equivalences to G -equivalence. The homotopy orbit and homotopy fixed points constructions therefore take underlying equivalences to equivalences of spaces, and Borel cohomology takes underlying equivalences to isos.

Then

For any G -space X and abelian group A of coefficients, the groups

$H_{\text{Borel}}^*(X)$, as H ranges over the subgroups of G , form a G -Mackey functor.

Denote $H_{\text{Mack}}^*(X)$ as the Mackey functor.

Def (cohomological Mackey functor)

$\underline{M} \in \text{Mack}_G$ is a cohomology Mackey functor, if for each $H \leq K$, the composition

$$\underline{M}(K) \xrightarrow{R_H^K} \underline{M}(H) \xrightarrow{I_H^K} \underline{M}(K)$$

is multiplication by the index $|K:H|$.

Def (orbit category)

Let $\mathcal{O}_G$ denote the orbit category of G , whose objects are the G orbits G/H and whose morphisms are just the G -equiv maps.

$\mathcal{O}_G$ is a full subcategory of the category FinSet_G of finite G -sets. The set $\text{Hom}_{\mathcal{O}_G}(G/H, G/K)$ is nonempty iff H is contained in some conjugate of K .

Def (coefficient system)

A coefficient system for G is a functor $M: \mathcal{O}_G^{\text{op}} \rightarrow \text{Abgrp}$.

Ex
For $n \geq 2$, G -space X , $\underline{\pi}_n(X)(G/H) := \pi_n(X^H)$ defines a coefficient system.

Ex

G -space X , abelian group A , $n \geq 0$. $H_n(X; A)(G/K) = H_n(X^K; A)$ defines a coefficient system.

Def

A map of G -coefficient system $C \rightarrow D$ is a natural transformation of functors. Write $\text{Hom}_{\text{coeff}}(C, D)$ for the set of maps of coefficient systems.

Ex

$C_*(X)(G/K) = C_*^{\text{cell}}(X^K)$ is a chain complex. The differentials in these complexes commute with the restrictions and conjugation homos.

$$\begin{array}{ccc} C_n^{\text{cell}}(X^K) & \xrightarrow{d_n} & C_{n-1}^{\text{cell}}(X^K) \\ \downarrow R_H^K & & \downarrow R_H^K \\ C_n^{\text{cell}}(X^H) & \xrightarrow{d_n} & C_{n-1}^{\text{cell}}(X^H) \end{array}$$

So $d_n: C_n(X) \rightarrow C_{n-1}(X)$ makes $C_*(X)$ a chain complex of coefficient systems.

Def (Bredon cohomology)

X is a G -space, and M is a G -coefficient system. $C_{\text{coeff}}^*(X; M) := \text{Hom}_{\text{coeff}}(C_*(X), M)$ is a cochain of abelian groups. We can

define the Bredon cohomology of X with coefficients M to be

$$H_{\text{Bredon}}^n(X; M) = H^n(C_{\text{Coeff}}^*(X; M)).$$

Usually write $H_G^*(X; M)$ for Bredon cohomology.

Prop

C is a G -coefficient system. Then, $\text{Hom}_{\text{Coeff}}(C, \underline{\mathbb{Z}}) \cong \text{Hom}_{\text{Ably}}(C(G), \mathbb{Z})$.

Ex

$G = C_2$. $X = S^\sigma$. $C_*(S^\sigma)$ is

$$\begin{array}{ccc} 0 & \longrightarrow & \mathbb{Z}^2 \\ \downarrow & & \downarrow \text{id} \\ \mathbb{Z}[C_2] & \xrightarrow{[1, -1]} & \mathbb{Z}^2 \end{array}$$

According to the Prop.,

$$\mathbb{Z} \xleftarrow{[1, -1]} \mathbb{Z}^2$$

So

$$H_{C_2}^k(S^\sigma; \underline{\mathbb{Z}}) \cong \begin{cases} \mathbb{Z} & k=0 \\ 0 & k>0 \end{cases}$$

Prop

For any G -space X , $H_G^n(X; \underline{\mathbb{Z}}) \cong H^n(X/G; \mathbb{Z})$

Prop

C is a G -coefficient system. $\text{Hom}_{\text{Coeff}}(C, \varinjlim_{G/G} \mathbb{Z}) \cong \text{Hom}_{\text{Ably}}(C(G), \mathbb{Z})$.

Prop

For any G -space X , $H_G^n(X; \inf_{G/G}^G(\mathbb{Z})) \cong H^n(X^G; \mathbb{Z})$.

$$\text{Hom}_{\text{coff}}(C, \uparrow_e^G(\mathbb{Z})) \cong \text{Hom}_{\text{Ab}G}(C(e), \mathbb{Z})$$

$$H_G^n(X; \uparrow_e^G(\mathbb{Z})) \cong H^n(X; \mathbb{Z}).$$

$$H_{C_p}^n(X; \mathbb{Z}) \cong H^n(X/C_p, X^G; \mathbb{Z}) \cong \tilde{H}^n((X/X^G)/C_p; \mathbb{Z})$$

Def

C is a G -coefficient system and N is a covariant $\mathcal{O}_G \xrightarrow{N} \text{Ab}G$.

Define $C \otimes_{\mathcal{O}_G} N \in \text{Ab}G$ as

$$C \otimes_{\mathcal{O}_G} N = \left(\bigoplus_{H \leq G} C(H) \otimes N(H) \right) / \sim$$

where

$$\tau(x) \otimes y \sim x \otimes \tau(y) \text{ for } x \in C(K), y \in N(H) \text{ and } H \leq K$$

$$C_g(x) \otimes y \sim x \otimes C_g(y) \text{ for } x \in C(K^g) \text{ and } y \in N(K).$$